Shivani Sareddy

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SUMMURY

Data Engineer with 4+ years of experience building and maintaining data pipelines and data systems using Python, SQL, Spark, and Kafka. Worked on batch and streaming data processing on AWS and Snowflake in financial services environments. Experienced in improving data quality, optimizing query and pipeline performance, and supporting reporting and analytics needs in production systems.

TECHNICAL SKILLS

Programming & Scripting: Python (PySpark, Pandas, NumPy, Scikit-learn), SQL (Advanced), Scala, Bash/Shell
Big Data & Data Engineering: Apache Spark, Kafka, Hadoop (HDFS, Hive), Airflow, Databricks (Delta Lake), dbt
Data Warehousing: Snowflake, Amazon Redshift, Google BigQuery; Dimensional Modeling (Star/Snowflake), Data Vault
Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Redshift, Kinesis), Azure (Data Factory, Synapse, Data Lake), GCP (BigQuery, Dataflow, Pub/Sub)
ETL / Data Integration: Informatica, Talend, API-based ingestion (REST/JSON, FIX), custom Python/Spark pipelines
AI / ML (Applied): ML pipelines (feature engineering, training datasets, deployment support), Scikit-learn, MLflow, Feature Stores, MLOps basics, batch/streaming data for ML use cases (fraud, risk, trading)
Financial Domain: Market data (tick, OHLC, order book), trading & risk systems, regulatory reporting (Basel III, MiFID II, SOX, GDPR), financial instruments (equities, derivatives, FX, fixed income), reference data (ISIN, CUSIP)
DevOps & Tools: Git, Jenkins, CI/CD, Docker, Kubernetes, Terraform, CloudFormation
Monitoring & Governance: Prometheus, Grafana, CloudWatch, ELK, Data Quality (Great Expectations), Data lineage (Collibra/Alation), RBAC/IAM

EDUCATION

Master's in computer science | Kent State University Kent, USA **August 2023 – December 2024**

EXPERIENCE

Data Engineer |Berkshire Hathaway, USA **April 2024 – Present**

- Engineered scalable enterprise data lakehouse architecture on AWS (S3, Glue, EMR, Redshift, Athena, Databricks Delta Lake) processing 250+ GB/day financial and trading data, improving analytics performance by 45% and reducing query latency significantly.
- Orchestrated real-time streaming pipelines using Apache Kafka and Spark Structured Streaming handling 5M+ events/second, enabling near real-time market data processing with ingestion latency under 10 seconds.
- Constructed high-performance ETL/ELT pipelines using PySpark, Python, and dbt, executing 3,000+ daily workflows, improving pipeline efficiency by 40% and reducing failure rate by 30% through optimization and partitioning strategies.
- Modeled enterprise-grade Dimensional Modeling and Data Vault architectures supporting Basel III, MiFID II, SOX regulatory reporting, processing 100+ reports/month with 100% audit traceability and 50% faster reconciliation cycles.
- Operationalized MLOps pipelines using MLflow, Python, and CI/CD automation, supporting 20+ ML models for fraud detection and risk analytics, reducing deployment cycle time from 14 days to 2 days.
- Governed enterprise data ecosystem using Collibra, Alation, IAM, RBAC, and encryption standards, managing 500+ datasets and 1,200+ data assets, improving data discoverability by 70% and reducing quality incidents by 35%

Data Engineer | Vivma Software Inc, India **March 2021 – July 2023**

- Developed high-performance PySpark-based ETL pipelines processing 2 TB/day transactional and application data, improving batch processing efficiency by 45% and achieving 99.5% SLA compliance.
- Optimized complex SQL workloads across Snowflake, PostgreSQL, and SQL Server, handling 500M+ records per table, improving query performance by 60% through indexing, partitioning, and query tuning techniques.
- Integrated scalable data ingestion systems using Apache Kafka, REST APIs, and batch processing frameworks, supporting 1M+ API transactions/day and enabling near real-time analytics with <30-second latency.
- Automated cloud-based workflows using AWS (S3, Glue, Lambda, Redshift), building 25+ production-grade pipelines, reducing manual intervention by 80% through orchestration and scheduling automation.
- Implemented enterprise data quality framework using Great Expectations, enforcing 200+ validation rules per pipeline, reducing downstream data errors by 40% and improving data reliability significantly.
- Streamlined CI/CD pipelines using Git, Jenkins, and Docker, deploying 15+ ETL releases/month, improving release stability by 50% and enabling rollback-based recovery strategy.

PROJECT

Real-Time Credit Risk Scoring Dashboard

Created a system to assess borrower risk instantly, combining several machine learning approaches to improve prediction results by 20%. Implemented a fast-scoring pipeline using Python and FastAPI to process tens of thousands of transactions per day efficiently. Produced interactive dashboards in Tableau and Plotly for clear visualization of portfolio performance and risk patterns. Used MLflow to monitor model performance and detect changes over time.

AI-Powered Fraud Detection System

Engineered a solution to identify fraudulent activity, significantly enhancing detection accuracy. Launched a microservice using Docker and FastAPI capable of handling high-volume real-time requests with minimal delay. Developed reporting dashboards in Power BI to track suspicious activity and support timely intervention. Helped prevent over a million dollars in potential losses annually through automated alerts